DATERA EDU: Requirements Traceability

The table links each group of requirements to the part of the system that implements it and to the way it is checked. Components are from the Component Diagram. Scenarios S1-S12 and the tests are from the Verification Plan. Requirement codes are from the technical specification.

Code	Requirement	Priority	Where it is implemented	How it is checked
FR-1	Registration, login, access control	MVP	Auth & access	S1, S7; functional checklist
FR-2	Consent to processing of recordings	MVP	Consent	S1, S9
FR-3	Subjects, materials and topics	MVP	Learning structure; Material topics (ML service)	S2
FR-4	Lesson recording upload	MVP	Lessons & upload; Pipeline orchestrator; Scheduler; MinIO	S3, S8
FR-5	Audio cleaning	MVP	ML service, stages 0-1	ML pipeline tests (error rate with and without cleaning); S3
FR-6	Speech recognition and speaker separation	MVP	ML service, stages 2-3	ML pipeline tests (WER, DER, teacher detection); S4
FR-7	AI analysis of the lesson	MVP	Anonymiser; LLM analysis (stage 4); LLM adapter; Cloud LLM API	Control set of 5-10 lessons (agreement with the teacher, valid JSON); S3
FR-8	Mastery index and knowledge map	MVP; FR-8.3 to 8.5 stage 2	Index & knowledge map	S5; control set; S11 for stage 2 parts
FR-9	Reports for teacher, student, parent	MVP; FR-9.5 to 9.7 stage 2	Report writer (stage 5); Reports & cabinet views	S6; S11 for the weekly summary
FR-10	Tests and question bank	Stage 2	Tests & question bank	S10
FR-11 to FR-13	Teacher, student and parent portals	MVP	Web client (cabinets)	S6, S7; end-to-end tests in the browser
FR-14	Notifications	Stage 2	Notifications & digests	S11
FR-15	System administration	MVP	Admin & monitoring	S9, S12; functional checklist
FR-16	Training data and fine-tuning	Trial in MVP; model in stage 2	Training data; Training GPU server; Local LLM server	S5, S12; comparison of base, fine-tuned and cloud models on the control set
NFR-1	Performance	MVP	Pipeline; Backend API; Web client	Timing and load tests (S3)
NFR-2	Reliability	MVP	RabbitMQ; Task worker; PostgreSQL backup	Restart and retry tests; one backup restore
NFR-3	Security	MVP	Auth & access; Nginx; MinIO	S7; security tests

Code	Requirement	Priority	Where it is implemented	How it is checked
NFR-4	Personal data	MVP	Consent; Scheduler; server in Kazakhstan	S8, S9; legal check of the consent text before the pilot
NFR-5	Quality of ML components	MVP; NFR-5.7 stage 2	ML service	ML pipeline tests; control set; pilot (NFR-5.8)
NFR-6	Compatibility and usability	MVP	Web client	Browser tests; usability test of the Figma prototype
NFR-7	Maintainability	MVP	Docker Compose; CI; Backend API	Unit tests on every change; coverage check (NFR-7.4)